

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Pipeline Equipment Data Sheets - Pig Launchers & Receivers
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-005
Document Revision	R01
Document Status	Issued For Review
Originator	C. Anthony
Security Classification	Restricted
Issue Date	9 Aug 2022

Pipeline Equipment Data Sheets - Pig Launchers & Receivers

R01	09.08.22	Issued For Review	C. Anthony	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

TABLE OF CONTENTS

1	Scope.....	3
1.1	Project Reference Documents.....	
1.2	Codes and Standards.....	
2	Abbreviations.....	
3	Important Notes.....	
4	Datasheet 1: DATASHEET FOR PIG LAUNCHER (BBOU-A-200) BBOU MANIFOLD PLATFORM	4
5	Datasheet 2: DATASHEET FOR PIG LAUNCHER (ELEP-A-200) ELEPA MANIFOLD PLATFORM.....	5
6	Datasheet 3: DATASHEET FOR PIG LAUNCHER (ELEP-A-300) ELEPA MANIFOLD PLATFORM.....	6
7	Datasheet 4: DATASHEET FOR PIG LAUNCHER (AKAB-A-100) AKABEL MANIFOLD PLATFORM.....	7

1. SCOPE

This document covers data sheets for the Pig launcher and Receivers to be used within BBOU manifold Station, Elepa Manifold Station and Akabel Pigging Station.
This document shall be read in conjunction with the project reference documents listed below.

1.1 PROJECT REFERENCE DOCUMENTS

S/No	Document Number	Description
1	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis for Design
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-011	Specifications for Pig Traps
3	EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-006	Pig Trap Calculation Note
4	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-003	Specifications for Mainline Valves
5	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets 1 - 8
6	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-012	Specifications for Pig Trap End Closure
7	EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-014	General Piping Arrangement Drawing - BBOU
8	EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-015	General Piping Arrangement Drawing - ELEP
9	EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-016	General Piping Arrangement Drawing - AKAB
10	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems

1.2 CODES AND STANDARDS

S/No	Document Number	Description
1	API 5L	Specification for Line Pipe
2	ASME B31.8	Gas Transmissions and Distribution Piping Systems
3	ASME B16-47	Large Diameter Steel Flanges NPS 26 Through NPS 60
4	ASME Sec. VIII div.1 & DIV. 2	Boiler And Pressure Vessel code
5	ASME Sec. V	Non-Destructive Examination

2. ABBREVIATIONS

Abbreviation	Definition
AED	Anti-Explosive Decompression
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
CA	Corrosion Allowance
CS	Carbon Steel
DPT	Dye Penetrant Test
EN	European Standards
GR	Grade
LCB	Low Temperature (Wrought) Carbon Grade B
LTCS	Low Temperature Carbon Steel
MPI	Magnetic Particle Inspection
MSS	Manufacturers Standardization Society
NDT	Non-Destructive Testing
RF	Raised Face
RTJ	Ring Type Joint
RPTFE	Reinforced Polytetrafluoroethylene
RT	Radiographic Testing
SW	Socket Welded
UT	Ultrasonic Testing
WCB	Wrought Carbon Grade B

3. IMPORTANT NOTES

- 3.1 This data sheet shall be completely filled by the vendor, stamped, signed and deviations clearly marked for acceptance. Quotation without this requirement will not be considered for technical review and evaluation.
- 3.2 The Quick Opening Closures (QOCs) shall be supplied with a Key Interlock Safeguarding Systems as indicated in the PIDs. Pig Trap Vendor shall be responsible for the procurement of all other valve interlock to be installed on other valves
- 3.3 The Pig Trap shall be sloped with a minimum of 3deg as shown on the sketches.

DATASHEET FOR PIG RECEIVER									
ELEP-A-200									
ELEPA MANIFOLD PLATFORM									
VESSEL DATA									
Equipment name	Pig Receiver		Equipment Tag		ELEP-A-200				
Manufacturer			PED category						
Location	Outside		ATEX zone						
Orientation	Horizontal		NACE MR0175/ISO 15156						
Design code	ASME B31.8								
MAIN DIMENSIONS									
Intelligent PIG possible	Yes								
Pressure rating	900 #								
Minor barrel diameter / wall thickness	24" / 15.88 mm								
Minor length	2946 mm								
Major barrel diameter / wall thickness	28" / 17.5 mm								
Major length	6600 mm								
Reducer length	610 mm								
Base to centre line (minor / major)	1200 mm / 1250 mm								
Weight empty / hydro test (approx.)	TBA kg / TBA kg								
CONDITIONS									
Product	Gas and Condensate (Multiphase)								
Sour service	No								
Operating pressure min.	barg		Design pressure		110 barg				
Operating pressure max.	100 barg		Design temperature min.		-29 °C				
Operating temperature min.	°C		Design temperature max.		100 °C				
Operating temperature max.	°C								
MATERIAL SPECIFICATIONS									
Shell & cone	API 5L X70 PSL 2		Bolt / nuts - external		ASTM A193-B7 and A194-2H				
Internal cladding	NO		Bolt / nuts - internal						
Liner thickness	mm		Gaskets		Octagonal Ring Joint, 900#, ASME B16.20, Soft face				
Flanges	A860 WPHY 70		Internals		welded				
Nozzle necks - pipe	API 5L X70 PSL 2				removable				
Nozzle loads	Vendor to advise		External welded attachments						
Forgings / fittings	A684 F65/A860 WPHY 70		Supports		All support fixed				
Reinforcing plates	Vendor to advise		Pickling / passivating / brushing						
Anchor bolts	To be Provided by Vendor		Shotblasting						
Corrosion allowance	3 mm		Internal coating system		N/A				
Corrosion allowance liner	mm		External paint system		YES				
TESTING AND INSPECTION									
Hydrostatic test pressure	138 barg								
Hydrostatic test temperature	°C								
PWHT due to service									
PWHT as per code									
Inspection witnessed by									
HIC testing	YES								
Impact testing requirement	NDE								
Material certificate	Type 3.2 (EN 10204)								
END CLOSURE									
End closure opening (Note 2)	Refer to EA8FPCLGPMC GGP/FEEDPPLSPC-012 Specifications for Pig Trap End Closure								
Number of pressure cycles (door openings)									
NOTES									
DESCRIPTION OF REVISIONS									
General: Pig Trap System design shall comply with ASME B 31.8 and DEP 31.40.10.13-Gen.									
Note 1: Refers to section numbers of DEP 31.40.10.13-Gen.									
Note 2: For hinged closures, the Company shall indicate the required opening orientation (right/left/up)									
Made by Date EQUIPMENT : Pig Receiver ELEP-A-200 Rev. letter									
E. Uwama 07/08/22 PLANT : ELEPA Manifold Platform Date									
Checked by Date E. Nwachukwu 07/08/22 Sign.									
Approved by Date CONSIGNEE : Brass Fertilizer and Petrochemical Company Limited (BFPCl) Sheet No. 1 continued on sheet No. 2									
A. Duru 07/08/22									
Eng. by									
Principal Equipm. No.									
Req. No.									

CONNECTION SCHEDULE PIG RECEIVER									
Nozzles	Applicable	Mark	Size [inch]	Rating / Facing	Orientation	Elevation	References, remarks		
Inlet / outlet		N1	24	900 #	Front				
Closure		M1	28	900 #	End				
Drain		N2	4	900 #	Bottom				
Drain		N3	4	900 #	Bottom				
Balance line		N4	4	900 #	Top				
Pressure Indicator		N5	2	900 #	Top				
PIG signaliser		N6	2	900 #	Top				
Relief / Blowdown line		N7	4	900 #	Top				
Vent / Purge		N8A	2	900 #	Top				
Vent / Purge		N8B	2	900 #	Top				
Kicker Line		N9	8	900 #	Top				
Pressure Indicator		N10	2	900 #	Top				
24" X 28" PIG RECEIVER.									
ELEP-A-200									
Max. 3"									
Min.									
2946									
610									
6600									
NOTES									
DESCRIPTION OF REVISIONS									

